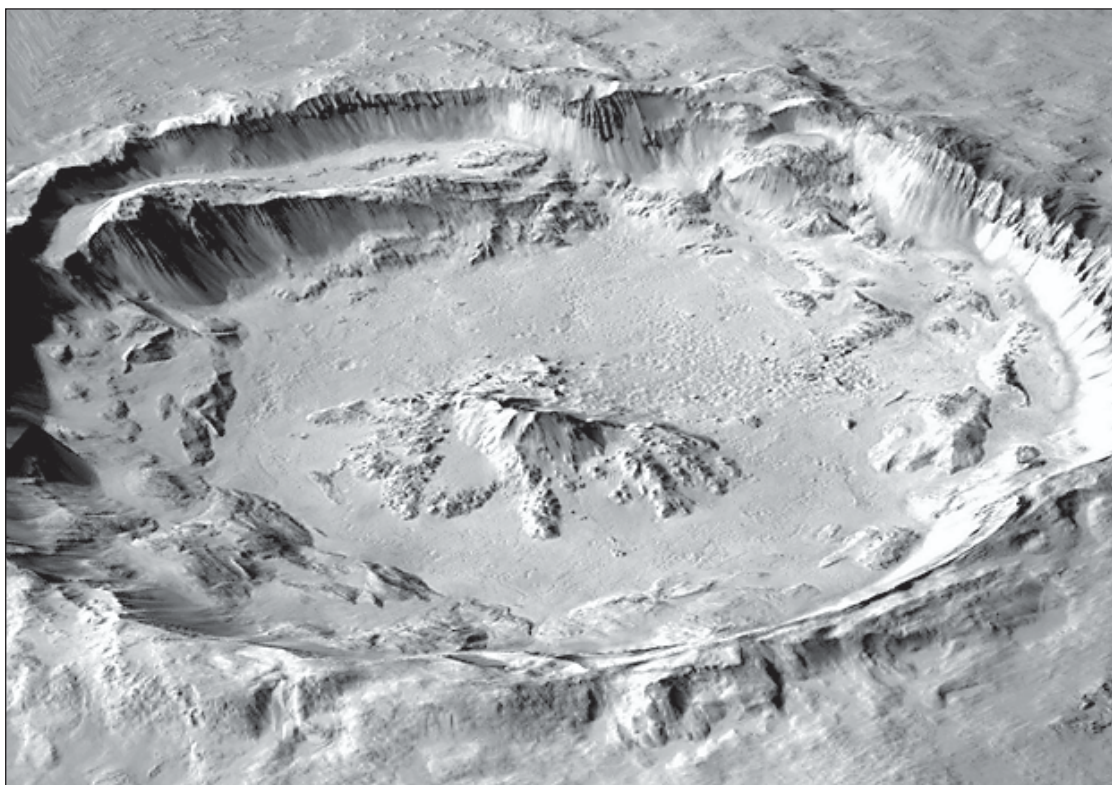


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Geologic Map of Tooting Crater, Amazonis Planitia Region of Mars

By Peter J. Mouginis-Mark

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Tooting crater has a diameter of 27.2 km and formed on virtually flat lava flows within Amazonis Planitia, Mars. The impact crater, which is located approximately 1,300 km west of the summit of Olympus Mons volcano, is very young and so provides a rare opportunity to map geomorphic units that are typically lost to erosion at older craters. Tooting crater is classified as a Multiple-Layered-Ejecta (MLE) crater.

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Introduction

Since the earliest days of the Viking Orbiter missions, analysis of lobate deposits surrounding many Martian impact craters suggested the presence of water or ice within the top kilometer or more of the crust at the time of crater formation (Carr and others, 1977; Gault and Greeley, 1978; Mouginis-Mark, 1979). The terms “rampart crater” (Mouginis-Mark, 1978) and “single-layered”, “double-layered”, and “multi-layered ejecta craters” (Barlow and others, 2000) have been used to describe these craters. Details of the ejecta emplacement process may provide an understanding of the rheology of the flows and the flow process (Woronow, 1981; Baloga and others, 2005), as well as provide information on the possible interactions of target volatiles with impact melt generated during the event (Kieffer and Simonds, 1980; Mouginis-Mark, 1987). By virtue of its excellent degree of preservation and extensive coverage by high-resolution images, Tooting crater provides an outstanding opportunity to explore these relations.

Tooting crater (23°10' N., -152°10' E.) formed within unit Aa₃, interpreted by Scott and Tanaka (1986) as Late Amazonian lava flows. Tooting crater is very young, as is evident from the lack of large superposed impact craters on the ejecta layers. Using images from the High Resolution Imaging Science Experiment (HiRISE), Mouginis-Mark and Boyce (2012) made crater counts for several of the ejecta layers (units el₃, el₂, and el₁) and derived an age for the Tooting impact of ~2.9±0.3 Ma. The geologic units within which Tooting crater formed (described here as “target material”) also has a young age, estimated to be 240 to 375 Ma (Mouginis-Mark and Boyce, 2012), raising the possibility that Tooting crater may be the source crater for some of the basaltic shergottite meteorites, which have ejection ages of 2.4±0.2 Ma to 3.0±0.3 Ma and crystallization ages from ~165±11 Ma to 327±10 Ma (McSween, 2008).

Tooting crater has a diameter of 27.2 km and formed on virtually flat lava flows within Amazonis Planitia ~1,300 km west of the summit of Olympus Mons volcano (fig. 1), where there appear to have been no other major topographic features prior to the impact. The crater formed in an area ~185 x 135 km that is at an elevation between -3,870 m and -3,874 m relative to the Mars Orbiter Laser Altimeter (MOLA) Mars datum (Mouginis-Mark and Garbeil, 2007). This fortuitous situation (for example, a bland, horizontal target) allows the geometry of the crater and the thickness of the ejecta blanket to be accurately determined by subtracting the appropriate elevation of the surrounding landscape (-3,872 m) from the individual MOLA measurements across the crater. Thus, for the first time, it is possible to determine the radial decrease of ejecta thickness as a function of distance away from the rim crest (Mouginis-Mark and Garbeil, 2007). On the basis of the four discrete ejecta layers surrounding the crater cavity, Tooting crater is classified as a Multiple-Layered-Ejecta (MLE) crater (Barlow and others, 2000). By virtue of the asymmetric distribution of secondary craters and the greater thickness of ejecta to the

northeast, Morris and others (2010) proposed that Tooting crater formed by an oblique impact from the southwest. The maximum range of blocks that produced identifiable secondary craters is ~500 km (~36.0 crater radii) from the northeast rim crest. In contrast, secondary craters are only identifiable ~215 km (15.8 radii) to the southeast and 225 km (16.5 radii) to the west.

Origin of Crater Name

Tooting crater was formally named in 2005 by the International Astronomical Union. This name derives from the town of Tooting in South London, England. Tooting was first described back in the year 675 A.D., when it was called “Totinge.” It was also mentioned in the Domesday Book in 1086. “Tooting” comes from the name of the Saxon Tota and follows the old English custom of adding “ing” (which means “the people who lived at”) to the person’s name. Thus “Tooting” literally means “The home of Tota’s people.”

Data

The primary data set used for the mapping came from the Thermal Emission Imaging System (THEMIS) visible (VIS) camera (Christensen and others, 2004). THEMIS VIS data at 18 m/pixel cover almost all of the mapped area, with only a few data gaps (fig. 2A) which were filled with THEMIS infrared (IR) data with 100 m/pixel resolution. Additional consulted data sets (fig. 2B) comprised images from the Context Camera (CTX) (Malin and others, 2007) and HiRISE (McEwen and others, 2007) with a ground sample distance of ~6 m/pixel and ~25 cm/pixel, respectively. Table 1 lists the HiRISE images used for the supplemental mapping of Tooting crater. Topographic data were derived from a digital elevation model (DEM) produced from stereo CTX images (Mouginis-Mark and Boyce, 2012), controlled by MOLA spot elevation measurements (Smith and others, 2001).

Topography

Spot elevation measurements within Tooting crater (fig. 3, table 2) reveal that the maximum crater depth below the pre-impact surface, ~1,275 m, is on the southern floor of the crater. The northern floor has a depth of ~1,080 m. The minimum rim elevation is 347 m above the pre-impact surface, and the maximum elevation is 951 m. Thus the maximum depth/diameter ratio of Tooting crater, as measured from the highest point on the rim to the lowest point of the floor, is ~0.082. The minimum depth/diameter ratio is 0.039. The highest point on the central peak (unit cpm) is 1,110 m above the surrounding floor material, or 119 m below the pre-existing level of the target material and 768 m below the average rim crest elevation.